

Predicted

	Negative	Positive
Actual Negative	True negative	False positive
Actual Positive	False negative	True positive

Ground truth

$m = \#$ of samples

Example: classifying images based on whether they contain humans.

$$m = m_p + m_o$$

$\downarrow$ people $\downarrow$ other

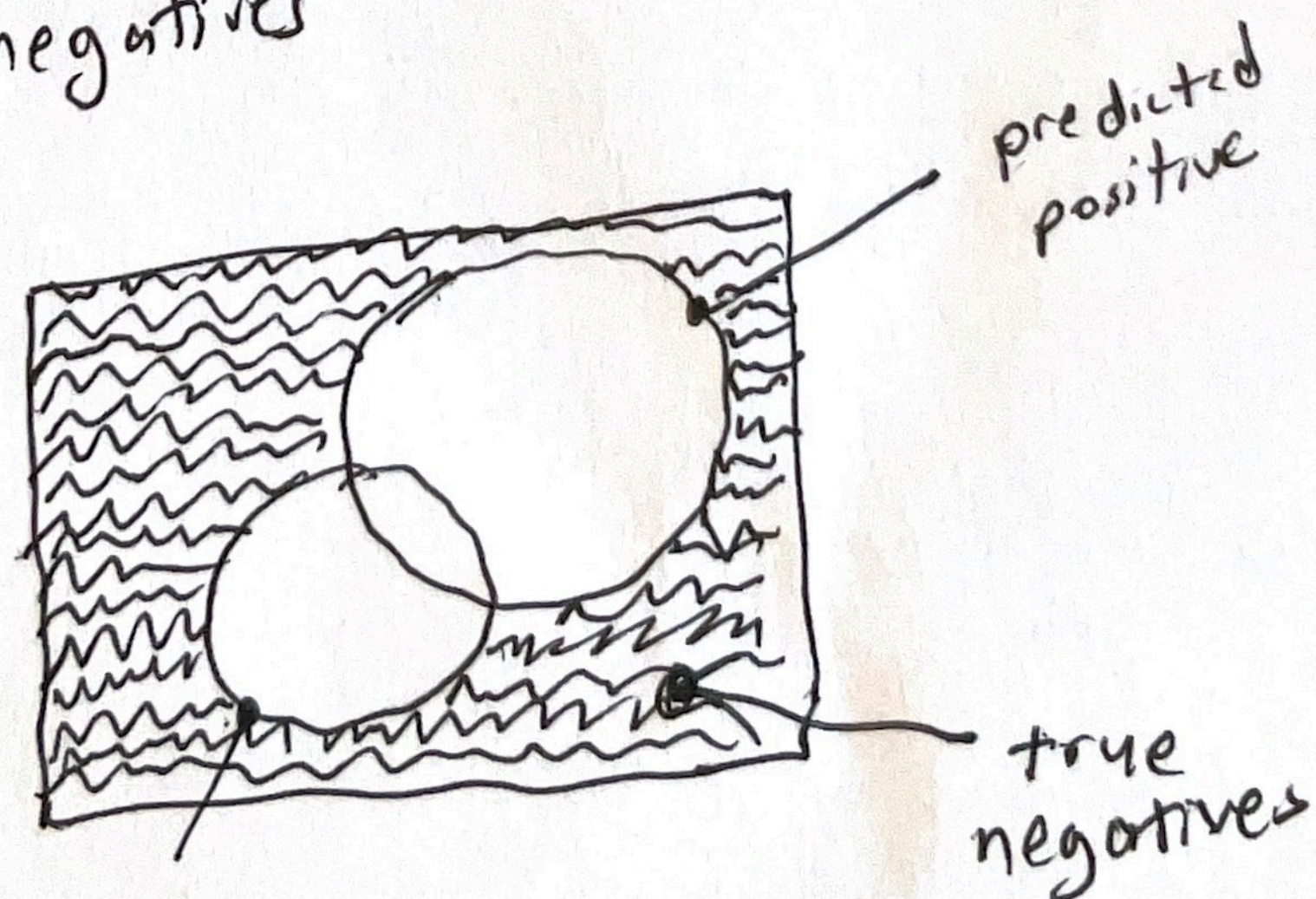
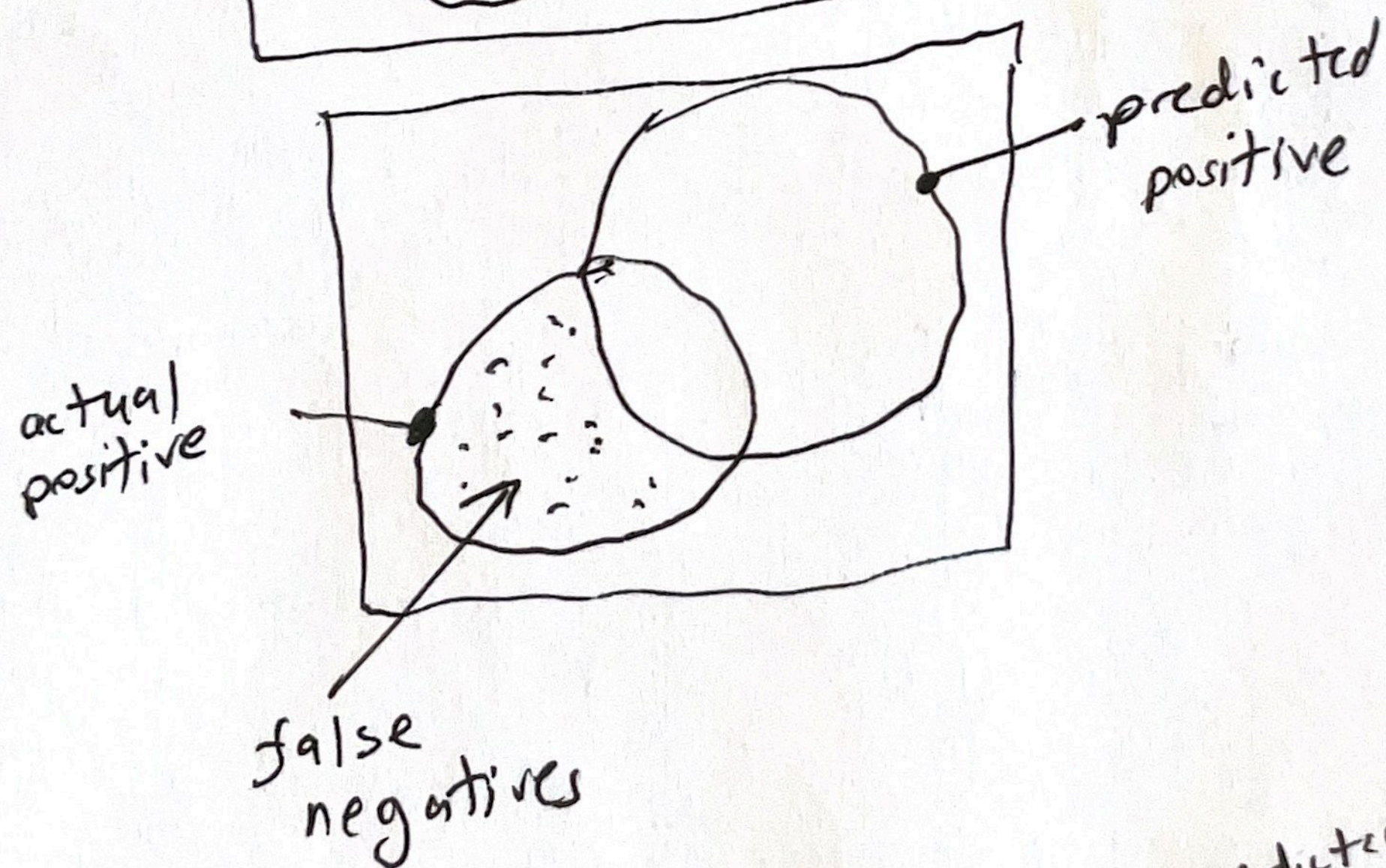
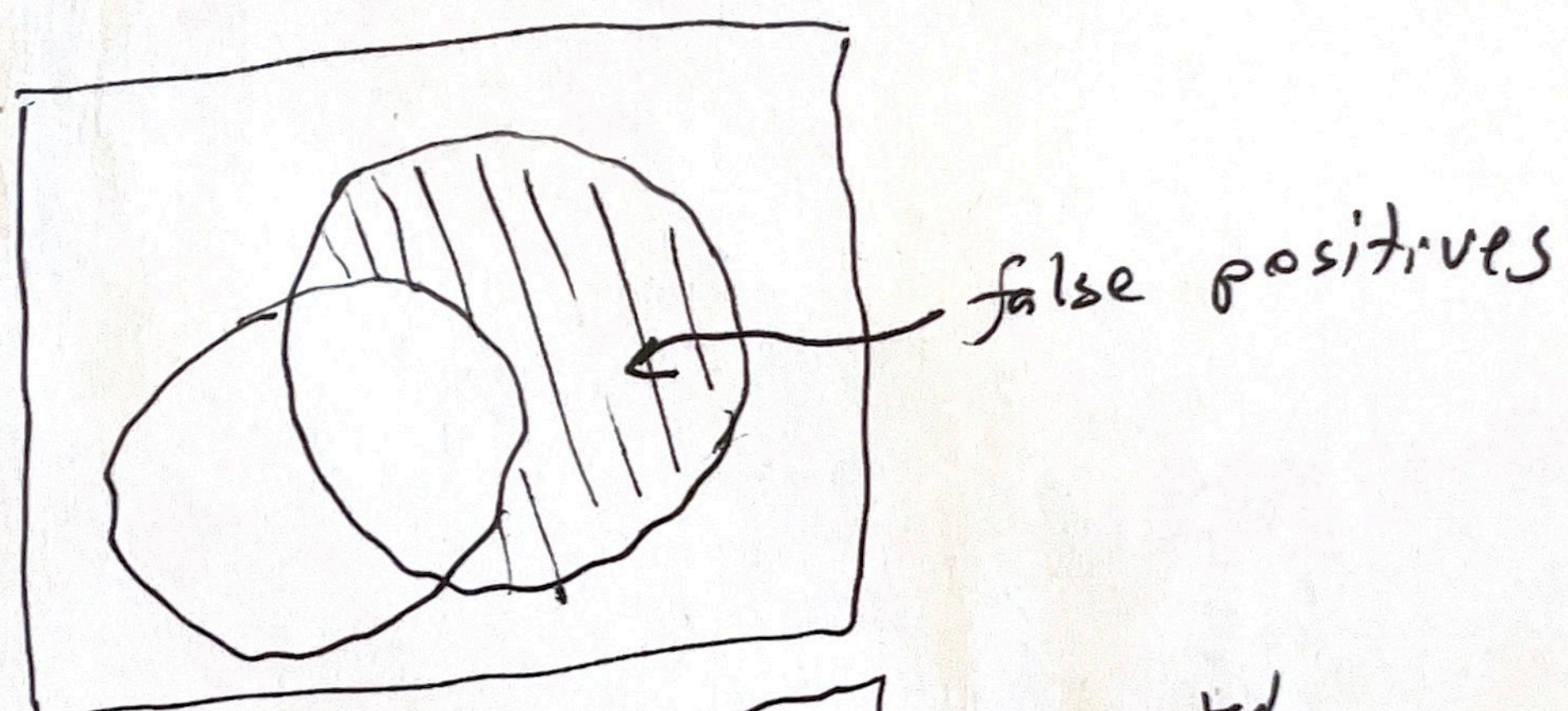
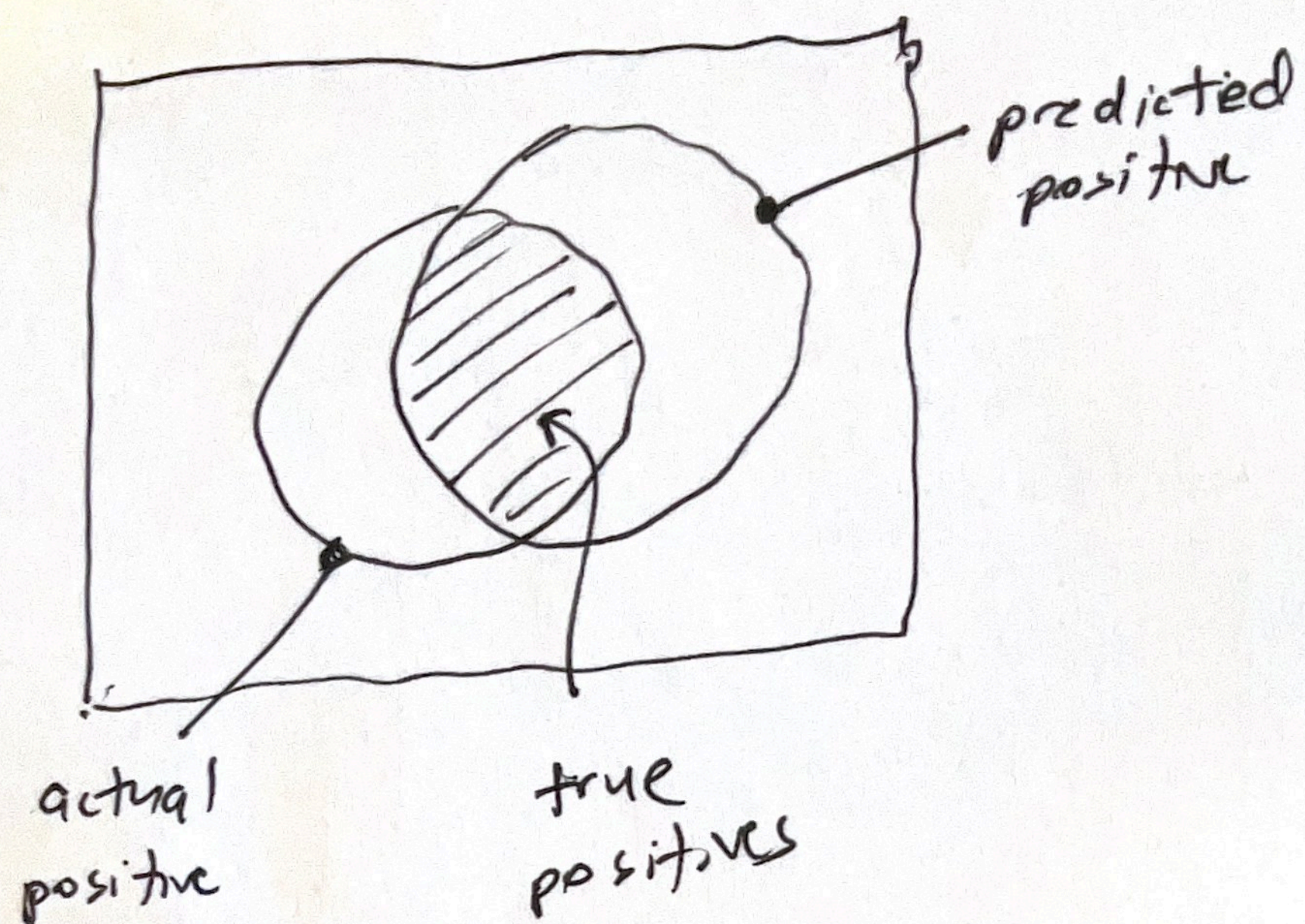
$$\text{accuracy} = \frac{TP + TN}{m}$$

$$\text{accuracy} = 1 - \text{error}$$

$$\text{precision} = \frac{TP}{TP + FP} = \frac{\text{number actually positive of predicted positive}}{\text{number predicted positive}}$$

positive
 $\downarrow$

$$\text{recall} = \frac{TP}{m_{\text{people}}} = \frac{\text{number actual positives of predicted positive}}{\text{number of actual positives.}}$$



Examples where we might intentionally bias toward precision

- Financial Fraud Detection
 - Don't frequently flag legitimate transactions.
 - When flagged, it is most likely fraudulent
 - A few minor acts of fraud are tolerable.

- Spam detection

- Don't filter potentially legitimate email
- A little spam leaking through is acceptable

- Jury Decisions

- Better to let a few guilty go unpunished than to jail the innocent (?)

Example of bias toward recall

- medical screening

- better to detect cancer and risk a few false positives

rather than allowing cancer to go untreated

- search engines / information retrieval

- better to include a few irrelevant results than to miss the desired information.

- anomaly detection in manufacturing

- better to throw out possibly good parts than use bad parts to build a plane.

Common Failure Modes: Data Imbalance

$$\text{Accuracy} = \frac{\# \text{ correct}}{\# \text{ sampled}} = \frac{TP + TN}{m}$$

What if I have few samples of true positives?

$$\text{If } m_- = 1000, m_+ = 10$$

Declare all negative!

$$\text{accuracy} = \frac{1000}{1010} \approx 99\%$$

When data is imbalanced perhaps use a metric that is small whenever precision or recall are low and high when both are high

$$F_B = (1 + B^2) \frac{\text{precision} \cdot \text{recall}}{B^2 \cdot \text{precision} + \text{recall}}$$

Most often used is $B = 1$

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

precision is 0? $F_1 = 0$

recall is 0? $F_1 = 0$

precision is 0.9 & recall 0.9

$$\frac{2 \cdot 0.81}{0.9 + 0.9} = 0.9$$

Example above

precision = 0, recall = 0, $F_1 = 0$, accuracy = 0.99

Declare all positive!

$$\text{accuracy} = \frac{10}{1010} \approx 0.01$$

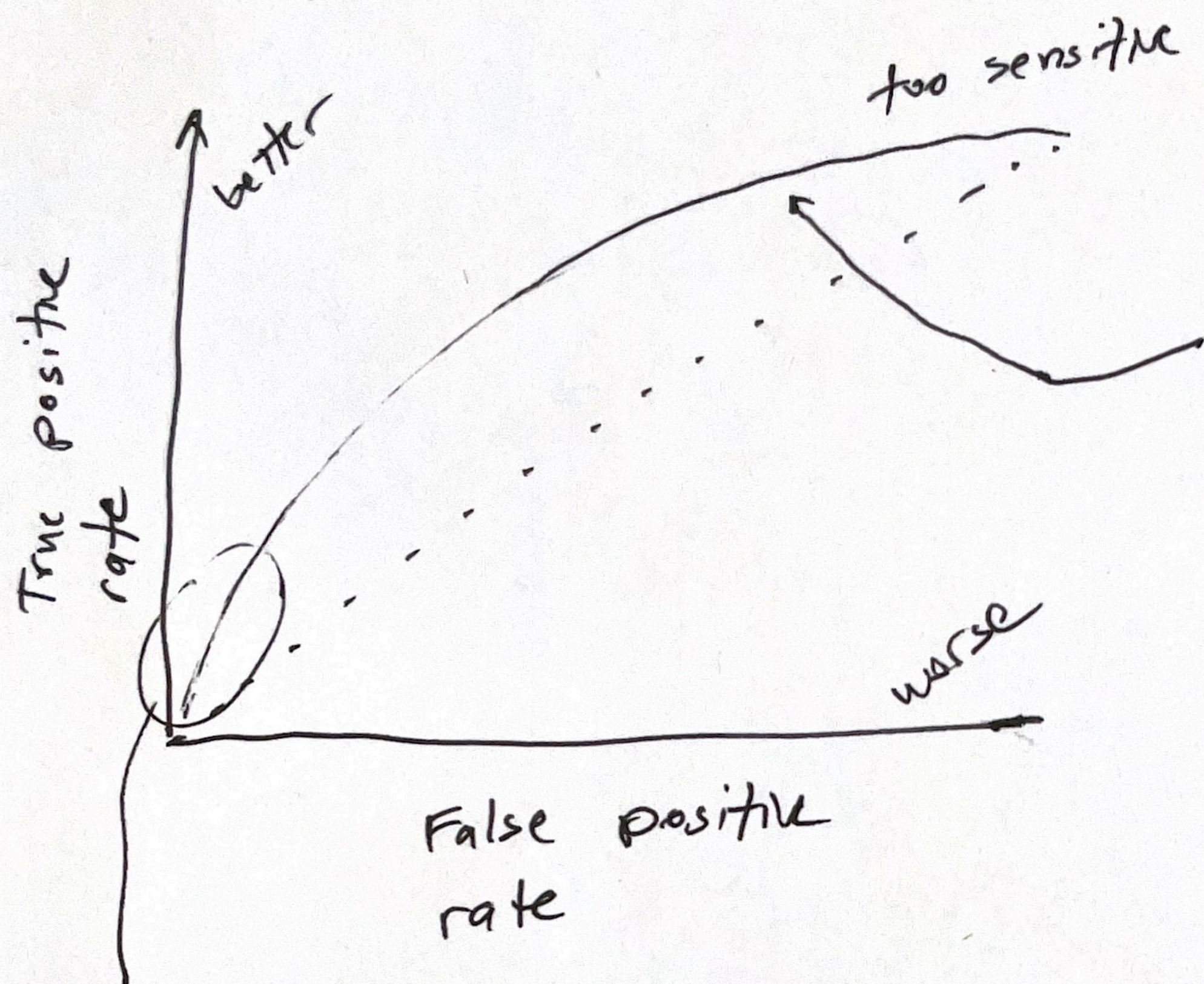
$$\text{precision} = \frac{10}{1010} \approx 0.01$$

$$\text{recall} = \frac{10}{10} = 1$$

$$F_1 = 2 \frac{0.01 \cdot 1}{0.01 + 1} \approx 0.02$$

ROC curves

"Receiver Operating Characteristic"



results as we vary a parameter, e.g., a detection threshold

too insensitive = hardly ever triggers for either actual positives or actual negatives

